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Yield-Aware Generative Inverse Design of Anti-Reflection Coatings via Optimal-Transport Flow Matching with a Conditional-Value-at-Risk Objective

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Abstract

Robustness to deposition error is conventionally treated as a post-hoc Monte-Carlo check in multilayer anti-reflection (AR) coating design: a nominally optimal stack is found first, and its manufacturing tolerance is assessed afterwards. We move robustness directly into the design objective. Instead of minimizing the mean reflectance, we minimize the conditional value-at-risk (CVaR) of the band-mean reflectance under the distribution of deposition errors, so that the optimizer is driven by the worst tail of likely fabrication outcomes rather than by an idealized nominal spectrum. This yield-aware formulation is combined with an Optimal-Transport Conditional Flow Matching (OT-CFM) generator operating under a hard manufacturability constraint enforced by a smooth box reparameterization, and with a differentiable transfer-matrix forward model validated against the analytic quarter-wave result to machine precision. On a six-layer MgF_2/SiO_2 stack evaluated on 10^5 held-out deposition realizations applied identically to every design, the CVaR-robust design lowers the 99th-percentile band-mean reflectance from 1.825 % to 1.799 %, an absolute reduction of 0.0257 pp (95% paired-bootstrap confidence interval $[+0.0230, +0.0274]$ pp; +1.41% relative, CI $[+1.26, +1.50]$%), at the cost of a 0.011 pp increase in nominal mean reflectance. Because both designs are scored on the same realizations, this is a paired comparison in which the shared sampling noise cancels; the improvement is small in absolute terms, and we report it together with the sampling uncertainty that establishes it. Repeating the entire optimize-then-evaluate workflow under five independent random seeds reproduces the effect in every run ($+1.22 \pm 0.10$%, mean ± s.e.m.). A deposition-error sweep shows the advantage is absent at $\sigma = 4$ nm, opens above that break-even point, and then plateaus: the *absolute* tail reduction increases monotonically across the sweep, whereas

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the *relative* gain rises to +2.34% at $\sigma = 12\,\text{nm}$ and settles at +2.19% at 16 nm. A material-choice ablation, restricted to the three six-layer recipes tested here on fused silica over 430 nm to 780 nm, documents a negative result: within this design space, introducing high-index TiO_2 degrades AR performance.

Keywords: inverse design; anti-reflection coatings; conditional value-at-risk; flow matching; fabrication tolerance; differentiable optics

1 Introduction

Multilayer anti-reflection coatings are among the most established components in optics [1, 7], yet their design still separates two questions that are physically coupled. First one asks which stack minimizes reflectance under ideal conditions; only then does one ask how much that performance degrades when the deposited thicknesses deviate from their targets. The second question is typically answered by a Monte-Carlo tolerance analysis performed *after* the design is fixed [3, 2]. If the answer is unsatisfactory, the designer returns to the first step and tries again. This holds for the classical synthesis methods, from automatic synthesis [5, 6] to needle optimization [4], as well as for the deep-learning approaches that have largely displaced them.

This separation is convenient but not obligatory. In integrated photonics, where lithographic variation can dominate device performance, variation-aware and yield-driven objectives have become standard: robustness enters the optimization itself rather than a subsequent audit [21, 22, 25, 24, 23]. The same logic applies to thin films. A deposition process has a characteristic thickness error, that error induces a distribution of reflectance spectra, and it is the upper tail of that distribution, not its nominal value, that determines manufacturing yield.

Deep learning has become a standard tool for photonic and thin-film inverse design [8, 9, 10, 11], with generative approaches [12, 13], autoregressive models [15, 14], adjoint-based methods [16] and differentiable transfer-matrix solvers [18, 19, 20] all applied to multilayer coatings; recent reviews survey the field [17]. What these share is a nominal objective: the target is an ideal spectrum, and fabrication tolerance is assessed afterwards if at all.

We adopt the conditional value-at-risk (CVaR), the mean of the worst $(1 - \alpha)$ fraction of outcomes, as the design objective [26, 27]. CVaR is coherent, convex in the loss, and admits the Rockafellar–Uryasev variational form that makes it differentiable and therefore usable inside a gradient-based inverse-design loop. It has been applied to risk-averse PDE-constrained optimization [28] and to certifiable engineering design [29], but not, to our knowledge, to the generative inverse design of optical coatings.

The generative component is an Optimal-Transport Conditional Flow Matching model [33, 34]. Flow matching learns a velocity field transporting a simple prior to a target distribution by regression on conditional paths, avoiding the simulation-based training of continuous normalizing flows; the minibatch optimal-transport coupling straightens those paths and improves sample quality at a given number of integration steps. Our interest in a generative model is not novelty for its own sake: a manufacturer rarely wants a single optimum, but a family of designs among which to choose on grounds the objective does not encode. Manufacturability is imposed as a hard constraint through a smooth box reparameterization,

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in the spirit of physics-constrained flow matching [35], so that no sampled design can violate the minimum-thickness limit.

This work extends our earlier post-hoc sensitivity and tolerance analysis of single-layer MgF_2 and SiO_2 coatings [41] from a diagnostic into a design objective for multilayer stacks.

Our contributions are: (i) a CVaR design objective for multilayer coatings, optimized end-to-end through a differentiable transfer-matrix solver; (ii) its combination with an OT-CFM generator under a hard manufacturability constraint; (iii) a paired, uncertainty-quantified evaluation protocol – bootstrap confidence intervals on every reported tail statistic and replication across independent optimization seeds – so that improvements of a few hundredths of a percentage point can be assessed against Monte-Carlo error rather than asserted; and (iv) an honest accounting of where the approach helps and where it does not, including a documented negative result on material choice.

2 Methods

2.1 Differentiable transfer-matrix forward model

For a stack of L homogeneous layers at normal incidence, the characteristic matrix of layer j at wavelength λ is

$$M_j(\lambda) = \begin{pmatrix} \cos\delta_j & \dfrac{i}{\tilde{n}_j}\sin\delta_j \\ i\,\tilde{n}_j \sin\delta_j & \cos\delta_j \end{pmatrix}, \qquad \delta_j = \frac{2\pi \tilde{n}_j(\lambda)\, d_j}{\lambda}, \tag{1}$$

with $\tilde{n}_j = n_j - ik_j$ the complex refractive index and d_j the physical thickness [38]. The stack matrix is the ordered product $M = \prod_{j=1}^{L} M_j$, and the amplitude reflection coefficient follows from its entries and the incident and substrate admittances,

$$r = \frac{n_0(m_{11} + m_{12}n_s) - (m_{21} + m_{22}n_s)}{n_0(m_{11} + m_{12}n_s) + (m_{21} + m_{22}n_s)}, \qquad R = |r|^2. \tag{2}$$

The solver is implemented in double precision with complex arithmetic throughout, so absorbing layers are handled correctly, and it is written in a framework providing reverse-mode automatic differentiation [37]; the Jacobian $\partial R/\partial d_j$ is therefore exact to machine precision rather than a finite-difference approximation.

Optical constants are read from tabulated dispersion data [39, 40] rather than from analytic Sellmeier fits, and are linearly interpolated onto the design grid. The design band is automatically clipped to the wavelength range common to all materials, so no extrapolation occurs.

Validation. For a single quarter-wave layer of index n_1 on a substrate of index n_s the reflectance at the design wavelength is $R = [(n_0 n_s - n_1^2)/(n_0 n_s + n_1^2)]^2$. With the tabulated values $n_{MgF_2}(550\,\mathrm{nm}) = 1.3844$ and $n_{SiO_2}(550\,\mathrm{nm}) = 1.4599$, our solver returns $R = 0.0182901883$ against the analytic 0.0182901883, a discrepancy of 0 to the printed precision ($|\Delta R| < 10^{-15}$), and the gradient at the quarter-wave optimum is -1.5×10^{-19}, numerically zero as required at a stationary point.

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2.2 Design space and hard manufacturability constraint

Layers thinner than roughly 10 nm do not form continuous films under island-growth conditions, so a designed thickness below that limit is not merely suboptimal but unphysical. Rather than penalizing such designs, we make them unreachable. Each layer is parameterized by an unconstrained latent z_j mapped through a smooth box transform,

$$d_j = d_{\min} + (d_{\max} - d_{\min})\,\sigma(z_j), \qquad \sigma(z) = \frac{1}{1+e^{-z}}, \tag{3}$$

with $d_{\min} = 10\,\text{nm}$ and $d_{\max} = 220\,\text{nm}$. The constraint is satisfied by construction for every $z \in \mathbb{R}^L$, so the generator cannot emit an invalid design.

2.3 Deposition-error model

We model deposition error as independent additive Gaussian perturbations of the layer thicknesses,

$$\tilde{d}_j = d_j + \varepsilon_j, \qquad \varepsilon_j \sim \mathcal{N}(0, \sigma_{\text{dep}}^2), \tag{4}$$

where σ_{dep} denotes the standard deviation of the thickness error; the baseline value is $\sigma_{\text{dep}} = 8\,\text{nm}$, representative of a sputter process without in-situ optical monitoring.This is the single significant modelling assumption of the study. Real deposition error is correlated between layers and generally non-Gaussian; a physically derived error distribution from a growth model would be preferable, and we return to this in the limitations.

2.3.1 Treatment of the thickness bounds under perturbation

A perturbed thickness $d_j + \varepsilon_j$ can approach or cross the hard bounds of Eq. (3), which occurs for the thinnest layers once σ_{dep} reaches 12 nm to 16 nm. Simply clipping to $[d_{\min}, d_{\max}]$ places a finite atom of probability exactly on the bound: for a layer sitting near $d_{\min}$ at $\sigma_{\text{dep}} = 16\,\text{nm}$, clipping would be active in roughly 45% of realizations. This truncates precisely the tail that the risk objective targets, and it biases the worst-case statistic downward, since a clipped realization can never be worse than the bound.

We therefore fold (reflect) the perturbation back into the box, the standard measure-preserving treatment of a bounded random walk: a layer driven below $d_{\min}$ re-emerges above it by the same amount, so no probability mass is destroyed and no artificial atom is created. The identical rule is applied inside the training objective and in the held-out evaluation, so the quantity optimized and the quantity reported are computed under the same assumptions. Clipping and rejection-resampling variants are both implemented and selectable; we verified that the $\sigma_{\text{dep}} = 8\,\text{nm}$ conclusions are unchanged under all three. The fraction of realizations for which any layer reaches a bound is reported alongside every result (Tables 3 and 5) and remains below 1% throughout.

2.4 Risk-aware objective

Let $\bar{R}(d, \varepsilon)$ denote the band-mean reflectance of design d under perturbation ε. The nominal objective is $\mathbb{E}_\varepsilon[\bar{R}]$ evaluated at $\varepsilon = 0$. The conditional value-at-risk at level α is the mean of

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the worst $(1-\alpha)$ fraction of outcomes, which by the Rockafellar–Uryasev identity [26] can be written

$$\mathrm{CVaR}_\alpha[\bar{R}] = \min_{\tau\in\mathbb{R}} \left\{ \tau + \frac{1}{1-\alpha}\,\mathbb{E}_\varepsilon\big[(\bar{R}-\tau)_+\big] \right\}, \tag{5}$$

where $(x)_+ = \max(x, 0)$. In practice we estimate τ by the empirical α-quantile of a minibatch of M perturbation samples and evaluate the outer expression, which is differentiable in d almost everywhere. Setting $\alpha \to 0$ recovers the mean objective exactly, so a single knob interpolates between nominal and robust design. We use $\alpha = 0.90$ throughout, targeting the worst decile of deposition outcomes; this is a conventional choice in risk-averse engineering design [29], and it retains enough tail samples per minibatch ($0.1 \times M$) for a stable gradient estimate.

2.5 OT-CFM generator

Flow matching learns a time-dependent velocity field $v_\theta(z, t)$ that transports a prior $p_0 = \mathcal{N}(0, I)$ to a target p_1 by regressing on straight conditional paths $z_t = (1-t)z_0 + tz_1$, with regression target $z_1 - z_0$ [34]. The optimal-transport variant reorders the prior samples within each minibatch according to the squared-Euclidean OT coupling before forming the paths, which reduces path crossing and improves sample quality at a fixed integration budget [33]. Where an exact earth-mover solver is unavailable, the implementation falls back to a GPU-resident entropic Sinkhorn solver (100 iterations) with a greedy assignment step to resolve ties, so the optimal-transport coupling is always active and never silently degrades to plain flow matching; the run reported here used the Sinkhorn path. The target distribution is a bank of robust designs obtained by refining a large batch of latents under the CVaR objective, in the spirit of population-based generative optimization. Sampling integrates $\mathrm{d}z/\mathrm{d}t = v_\theta(z, t)$ from $t = 0$ to $t = 1$ with a fourth-order Runge–Kutta scheme, after which the box map of Eq. (3) returns physical thicknesses.

2.6 Evaluation protocol and uncertainty quantification

All designs are scored on the *same* deposition realizations, the variance-reduction technique known as common random numbers [43]: the evaluation random-number generator is re-seeded identically for each candidate, so the nominal, robust and generated stacks encounter an identical set of thickness perturbations. Every design comparison is therefore a paired comparison under common random numbers, and the shared realization noise cancels in the difference.

This matters because the tail differences of interest are of order 0.03 pp. Compared on independent draws, such a difference is of the same order as the Monte-Carlo scatter of each percentile taken separately and could not be resolved at any practical budget; the paired difference, by contrast, is resolved with an interval roughly two orders of magnitude narrower. We regard this as the essential methodological point for evaluating small robustness gains, and it is the reason the improvements reported below can be stated with confidence intervals rather than asserted.

Uncertainties on all reported percentiles, and on the paired differences between designs, are obtained by percentile bootstrap over the held-out realizations (2000 resamples, 95%

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level) [42]. For the paired differences we resample realization *indices* rather than the two samples independently, which preserves the pairing. Any difference whose bootstrap interval contains zero is flagged as not distinguishable from Monte-Carlo error and is not claimed as an improvement.

We deliberately quote no confidence interval for the worst-case metric. The maximum of a finite sample is a high-variance estimator whose bootstrap distribution is degenerate at the upper order statistics, so a bootstrap interval for it would be misleading. The worst case is reported as a coarse supporting indicator only, and no conclusion in this paper rests on it.

Finally, we replicate the entire optimize-then-evaluate workflow under five independent random seeds. Each seed re-initializes both point-design optimizations and draws a fresh held-out Monte-Carlo set of 5×10^4 realizations. The tail improvement is reported as a mean $\pm$ standard error across seeds, together with the number of seeds favouring the robust design, so that the effect is attributed to the objective rather than to one fortunate optimization run.

2.7 Hyperparameters and reproduction details

Table 1 lists every setting needed to reproduce the results. The point-design baselines use Adam [36] with 600 steps, 6 random restarts and learning rate 3×10^{-2}; the CVaR gradient at each step uses $M = 64$ perturbation samples. The target bank contains 512 designs refined for 80 steps with $M = 16$. The velocity network is a three-hidden-layer multilayer perceptron of width 256 with SiLU activations and a 64-dimensional sinusoidal time embedding, trained for 1500 steps at batch size 256 and learning rate 2×10^{-3}, with path noise 10^{-3}; sampling uses 100 RK4 steps. All computation is in double precision.

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Table 1: Complete hyperparameter record for the reported run. The same values are exported in machine-readable form together with the code.

Component	Setting	Value
Physics	layers L	6 (MgF_2/SiO_2 alternating)
	design band	430 nm to 780 nm, 81 points
	thickness bounds	10 nm to 220 nm
	deposition error σ_{dep}	8 nm (baseline)
	boundary rule	reflecting (folded)
Risk objective	CVaR level α	0.90
	MC samples per step M	64
Point design	optimizer / steps	Adam / 600
	restarts	6
	learning rate	3×10^{-2}
Target bank	bank size	512
	refine steps / M	80 / 16
	learning rate	5×10^{-2}
OT-CFM	velocity net	MLP, 3 hidden layers, width 256, SiLU
	time embedding	sinusoidal, dimension 64
	training steps / batch	1500 / 256
	learning rate / path noise	2×10^{-3} / 10^{-3}
	OT coupling	entropic Sinkhorn, 100 iterations
	sampler	RK4, 100 steps
Evaluation	held-out realizations	10^5
	reporting seed / selection seed	12345 / 777 (independent)
	bootstrap resamples / level	2000 / 95%
	replication seeds	5 (5×10^4 MC each)
	sweep values of σ_{dep}	4, 8, 12, 16 nm

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3 Results

3.1 Problem setup

We design a six-layer alternating MgF_2/SiO_2 stack on a fused-silica substrate over 430 nm to 780 nm sampled at 81 wavelengths. In this band all three tabulated materials are transparent to the precision of the data ($k_{\max} = 0$), so the reflectance-only objective is well posed and $T = 1 - R$. The mean indices over the band are $n_{MgF_2} = 1.384$, $n_{SiO_2} = 1.459$ and $n_{TiO_2} = 2.504$; TiO_2 is loaded for the material ablation of Sec. 3.9 but is not part of the working stack.

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3.2 Nominal and robust point designs

Two baselines are optimized from identical initializations. Baseline A minimizes the nominal band-mean reflectance ($\alpha \to 0$); Baseline B minimizes $\mathrm{CVaR}_{0.90}$ with $M = 64$ perturbation samples per gradient step. The resulting stacks, from the air side inward, are

$$d_{\text{nominal}} = (101.1,\ 95.8,\ 183.0,\ 121.2,\ 191.5,\ 96.9)\ \text{nm},$$
$$d_{\text{robust}} = (100.0,\ 98.0,\ 171.7,\ 142.0,\ 184.4,\ 116.7)\ \text{nm}.$$

The two are not small perturbations of one another: the robust design redistributes thickness among the inner layers, thickening layers 4 and 6 by 21 nm and 20 nm respectively while thinning layer 3. The complete thicknesses of all three designs discussed in this paper, including the generated stack of Sec. 3.7, are collected in Table 2 and are also released in machine-readable form, so that any transfer-matrix implementation can reproduce the spectra and statistics reported here without re-running the pipeline.

Table 2: Optimized layer thicknesses (nm), listed from the air side (L1) to the substrate (L6). Together with the tabulated optical constants, these values are sufficient to reproduce every spectrum and Monte-Carlo statistic reported in this paper.

Layer	Material	Nominal-optimal	CVaR-robust	Generated-best
L1	MgF_2	101.1	100.0	99.6
L2	SiO_2	95.8	98.0	98.8
L3	MgF_2	183.0	171.7	179.0
L4	SiO_2	121.2	142.0	138.8
L5	MgF_2	191.5	184.4	185.8
L6	SiO_2	96.9	116.7	26.7
Nominal mean R (%)		1.439	1.450	1.453
MC 99th percentile (%)		1.825	1.799	1.812

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3.3 Held-out Monte-Carlo comparison

Both designs are evaluated on 10^5 deposition realizations not used in training. As described in Sec. 2.6, the same realizations are applied to both designs, so the comparison is paired. Results are given in Table 3.

Moving robustness into the objective reduces the 99th-percentile band-mean reflectance from 1.825 % to 1.799 %: an absolute reduction of 0.0257 pp with 95% paired-bootstrap confidence interval $[+0.0230, +0.0274]$ pp, corresponding to $+1.41\%$ relative (CI $[+1.26, +1.50]\%$, bootstrap $p < 10^{-3}$). The 95th percentile improves by 0.0115 pp ($+0.69\%$, CI $[+0.61, +0.74]\%$). The worst-case reflectance falls from 2.450 % to 2.385 % ($+2.66\%$), reported without an interval for the reason given in Sec. 2.6. The price is a 0.011 pp increase in nominal mean reflectance, from 1.439 % to 1.450 %: the robust design is slightly worse in the ideal case and better in the tail, which is precisely the intended trade. We state plainly that these are small absolute improvements. What the paired protocol establishes is that they are larger than the Monte-Carlo sampling error of the evaluation, and the seed replication of Sec. 3.4 establishes

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that they are a property of the objective rather than of a single optimization run. Both statements are necessary; neither alone would justify the claim.

Table 3: Held-out Monte-Carlo comparison over 10^5 deposition realizations at $\sigma_{\mathrm{dep}} = 8\,\mathrm{nm}$. Both designs are evaluated on identical realizations, so the comparison is paired, and 95% paired-bootstrap confidence intervals are reported for the percentile improvements. The clamped-fraction row reports how often any layer reached a thickness bound. No interval is quoted for the worst case, which is a high-variance estimator.

Metric	**Nominal optimal**	**Robust (CVaR)**	**Paired gain (pp)**	**95% CI (pp)**
Nominal mean R (%), no noise	1.439	1.450	-0.011	(cost)
MC mean R (%)	1.519	1.522	-0.003	(cost)
MC 95th percentile (%)	1.680	1.669	$+0.0115$	$[+0.0102, +0.0125]$
MC 99th percentile (%)	**1.825**	**1.799**	**+0.0257**	**[+0.0230, +0.0274]**
MC worst case (%)	2.450	2.385	$+0.065$	—
Clamped fraction (%)	0.0035	0.0002		

3.4 Replication across independent optimization seeds

To establish that the tail improvement is a property of the objective rather than of one optimization trajectory, the complete workflow – both point-design optimizations and a fresh held-out Monte-Carlo evaluation – was repeated under five independent random seeds. Results are given in Table 4 and Fig. 1.

The robust design attains the lower 99th percentile in all five runs. The mean relative gain is $+1.22\%$ with a standard error of 0.10% across seeds (standard deviation 0.21%, range $[+0.85, +1.39]\%$); in absolute terms $+0.022 \pm 0.002$ pp. The 95th percentile improves in all five runs ($+0.57 \pm 0.04\%$), as does the worst case, though with the much larger scatter expected of that estimator ($+2.14 \pm 0.57\%$, range $[+0.10, +3.30]\%$). The seed-to-seed variation in the gain is comparable in magnitude to the gain's own bootstrap interval within a single run, which is the honest characterization: the effect is small, consistent in sign, and modest in size relative to run-to-run variability.

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Table 4: Replication of the full optimize-then-evaluate workflow across five independent random seeds (5×10^4 Monte-Carlo realizations each). Each seed re-initializes both point-design optimizations and draws a fresh held-out Monte-Carlo set.

	p99 (%)		p99 gain		p95 gain	Worst gain
Seed	**Nominal**	**Robust**	(%)	(pp)	(%)	(%)
1000	1.822	1.807	+0.85	+0.0155	+0.49	+0.10
1137	1.822	1.799	+1.28	+0.0232	+0.61	+3.30
1274	1.820	1.797	+1.26	+0.0229	+0.68	+1.95
1411	1.820	1.796	+1.34	+0.0244	+0.59	+2.26
1548	1.824	1.799	+1.39	+0.0253	+0.50	+3.11
Mean			**+1.22**	**+0.022**	**+0.57**	**+2.14**
s.e.m.			±0.10	±0.002	±0.04	±0.57
Seeds favouring the robust design: 5 / 5						

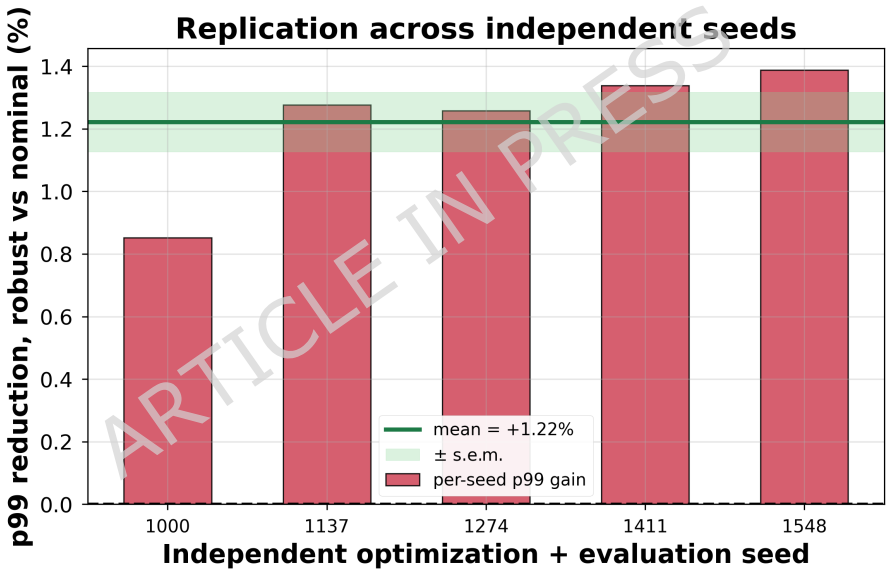


Figure 1: Per-seed 99th-percentile improvement across five independent optimize-then-evaluate replications, with the across-seed mean and standard error. The improvement is positive in every run.

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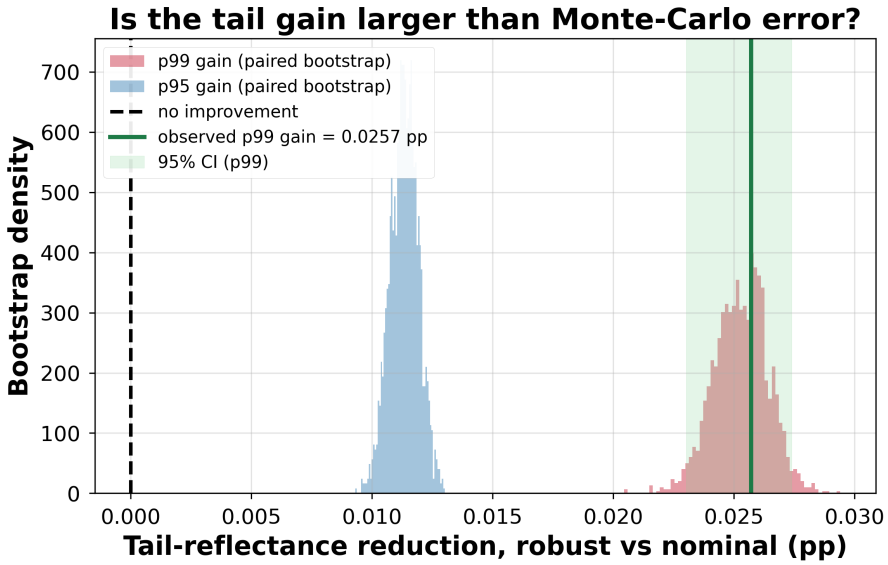


Figure 2: Bootstrap distribution of the paired 99th-percentile improvement over 10^5 held-out realizations (2000 resamples). The distribution is well separated from zero, which is what establishes that an improvement of 0.026 pp is resolvable at this Monte-Carlo budget.

3.5 Spectra and yield distributions

Figure 3 compares the two reflectance spectra. The nominal design achieves a slightly deeper minimum; the robust design trades depth for a flatter response, which is the spectral signature of tolerance to thickness error. Figure 4 shows the corresponding yield distributions under $\sigma_{\mathrm{dep}} = 8\,\mathrm{nm}$: the robust distribution is marginally shifted in the bulk but visibly lighter in the upper tail, where yield is decided.

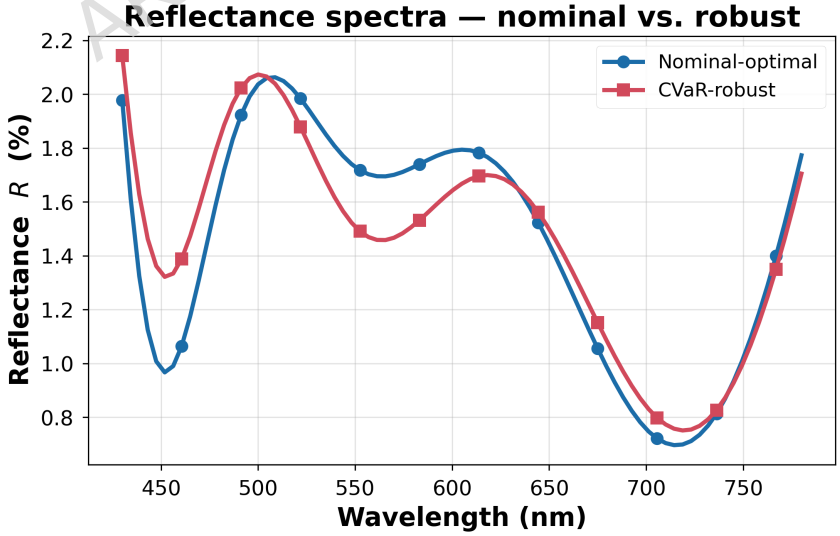


Figure 3: Reflectance spectra of the nominal-optimal and CVaR-robust six-layer designs over the design band.

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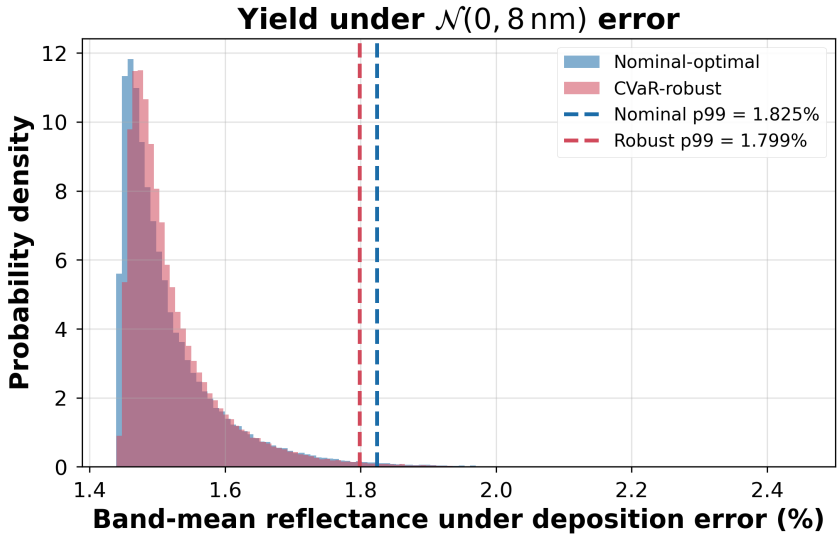


Figure 4: Distribution of band-mean reflectance over 10^5 deposition realizations under $\mathcal{N}(0, \sigma_{\mathrm{dep}}^2)$ deposition error with $\sigma_{\mathrm{dep}} = 8\,\mathrm{nm}$. Dashed lines mark the 99th percentiles.

3.6 Per-layer sensitivity

Because both designs sit at local optima, the first derivative $\partial \bar{R}/\partial d_j$ is approximately zero and carries no information about robustness. What distinguishes a tolerant design is the local *curvature*: how sharply reflectance rises under a thickness error of either sign. Figure 5 reports the mean upward change in band-mean reflectance when a single layer is displaced by $\pm\sigma_{\mathrm{dep}}$. The robust design shows a flatter curvature profile, with the reduction concentrated in the layers the CVaR objective chose to thicken.

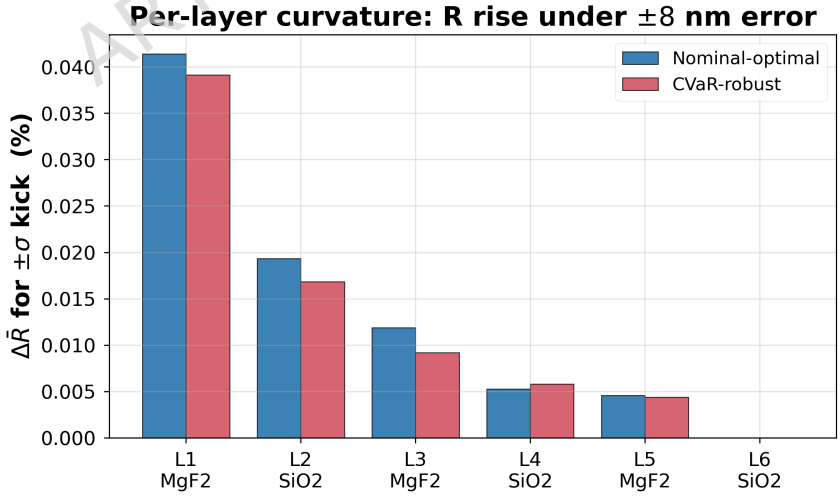


Figure 5: Per-layer curvature: mean rise in band-mean reflectance under a $\pm\sigma_{\mathrm{dep}}$ displacement of a single layer.

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3.7 Generative ensemble

The OT-CFM generator was trained on a bank of 512 CVaR-refined designs (median $\mathrm{CVaR}_{0.90} = 1.780\,\%$) and sampled 2000 designs; its training loss is shown in Fig. 6 and the per-layer diversity of the ensemble in Fig. 7. The generated ensemble has a median $\mathrm{CVaR}_{0.90}$ of 1.808 %, slightly above the bank it was trained on, as expected from a smooth generative model that regularizes the target manifold rather than memorizing it.

The best generated design was *selected* on an independent Monte-Carlo draw (seed 777) and then *reported* on the held-out draw (seed 12345) used for all other results. This separation matters: taking the minimum of 2000 noisy scores and quoting that same realization would bias the winner downward by selection optimism. Under this protocol the best generated design attains a 99th percentile of 1.812 %, between the nominal (1.825 %) and the directly optimized robust design (1.799 %), and the lowest worst case of the three (2.353 %). We note that the generator does not surpass direct CVaR optimization on the tail metric, and we do not claim that it should. Its value is that it supplies a diverse family of near-robust designs in a single forward pass, from which a manufacturer may select on criteria the objective does not encode. We also note that the best generated stack places its final layer at 26.7 nm, far thinner than the corresponding layer in either point design while still respecting the hard 10 nm constraint by construction; this illustrates that the generator explores genuinely distinct regions of the design space rather than reproducing the point optimum.

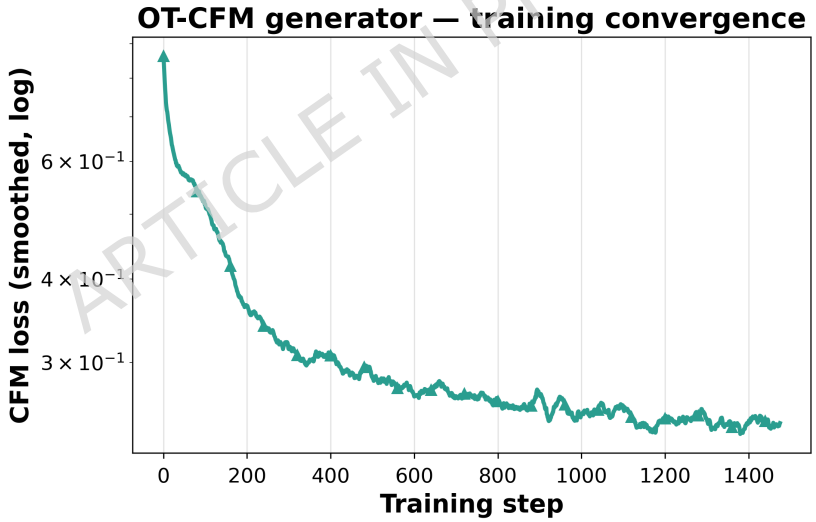


Figure 6: OT-CFM training convergence (smoothed, logarithmic axis).

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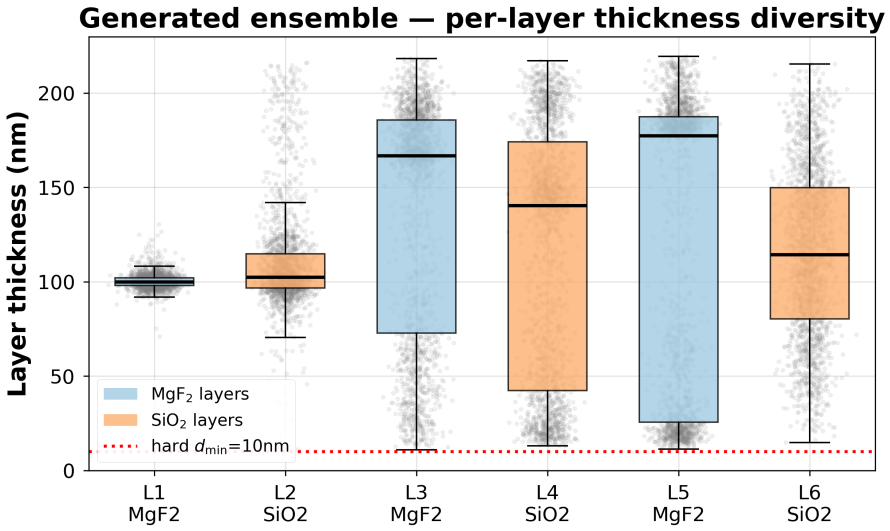


Figure 7: Per-layer thickness distribution of 2000 generated designs. The hard minimum-thickness constraint (dotted line) is never violated.

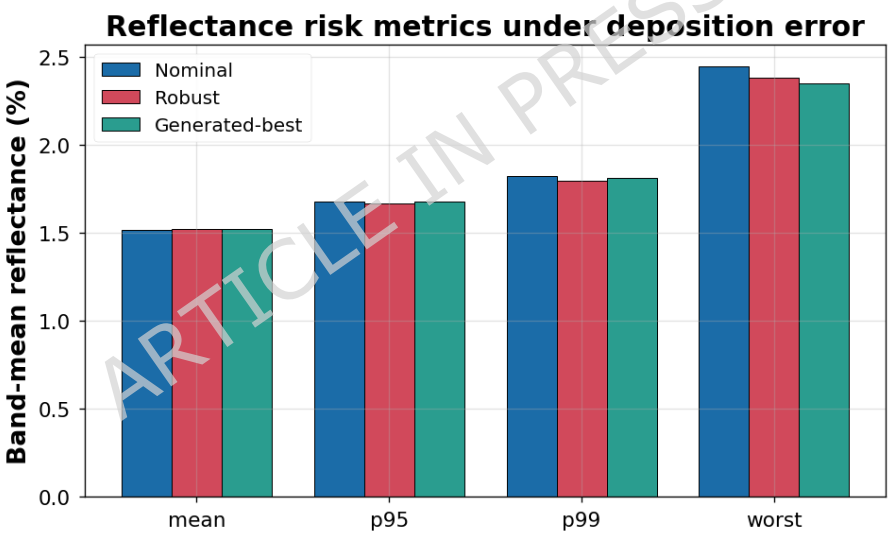


Figure 8: Risk metrics under deposition error for the three designs with 95% bootstrap confidence intervals shown as error bars. The designs are statistically indistinguishable in the mean and separate only in the tail.

3.8 Robustness across deposition-error magnitude

Table 5 and Fig. 9 report the comparison across four deposition-error magnitudes.

Two readings of the sweep must be distinguished, and we report both. The *absolute* 99th-percentile reduction increases monotonically with σ_{dep}: +0.0000, +0.0288, +0.0522 and 0.0585 pp at $\sigma_{\mathrm{dep}} = 4$, 8, 12 and 16 nm. The *relative* gain does not: it rises from 0.00% to +1.58% and +2.34%, then settles back to +2.19% at 16 nm, because the reference value $\bar{R}_{\mathrm{p99}}$ against which it is normalized is itself growing. We therefore describe the relative

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advantage as increasing and then plateauing, and we do not claim it improves monotonically. At $\sigma_{\mathrm{dep}} = 4\,\mathrm{nm}$ the two designs are statistically indistinguishable: the paired gain is $+0.00\%$ with confidence interval $[-0.11, +0.11]\%$, which contains zero. This is the break-even point of the method, and we report it as such rather than as a marginal win. Below roughly 4 nm of deposition error, robust design buys nothing in this system, and the 0.011 pp nominal penalty is a pure cost.

The worst-case column behaves less regularly, decreasing from $+2.37\%$ at $\sigma_{\mathrm{dep}} = 8\,\mathrm{nm}$ to $+1.14\%$ at 16 nm and turning slightly negative (-0.19%) at 4 nm. This is the expected behaviour of a single-realization estimator and is why we read the robustness benefit from the p99 trend and treat the worst case as supporting evidence only.

Table 5: Robustness sweep across deposition-error magnitudes, 10^5 realizations per entry. Absolute gains in percentage points and 95% bootstrap confidence intervals on the relative gains are reported alongside the relative figures, so that the non-monotonicity of the relative p99 gain and the sign change of the worst-case gain are visible in the table itself. Entries whose interval contains zero are marked †.

σ_{dep}	p99 (%)		p99 gain		Worst (%)		Worst	Clamp
(nm)	Nom.	Rob.	(%, 95% CI)	(pp)	Nom.	Rob.	gain (%)	(%)
4	1.544	1.544	$+0.00\,[-0.11, +0.11]^{\dagger}$	+0.0000	1.667	1.670	−0.19	0.00
8	1.829	1.801	$+1.58\,[+1.35, +1.77]$	+0.0288	2.246	2.193	+2.37	0.00
12	2.229	2.177	$+2.34\,[+2.07, +2.53]$	+0.0522	3.036	2.973	+2.06	0.16
16	2.671	2.612	$+2.19\,[+1.87, +2.37]$	+0.0585	3.920	3.875	+1.14	0.80

†Interval contains zero: not distinguishable from Monte-Carlo error. The absolute p99 gain increases monotonically with σ_{dep}; the relative p99 gain increases and then plateaus.

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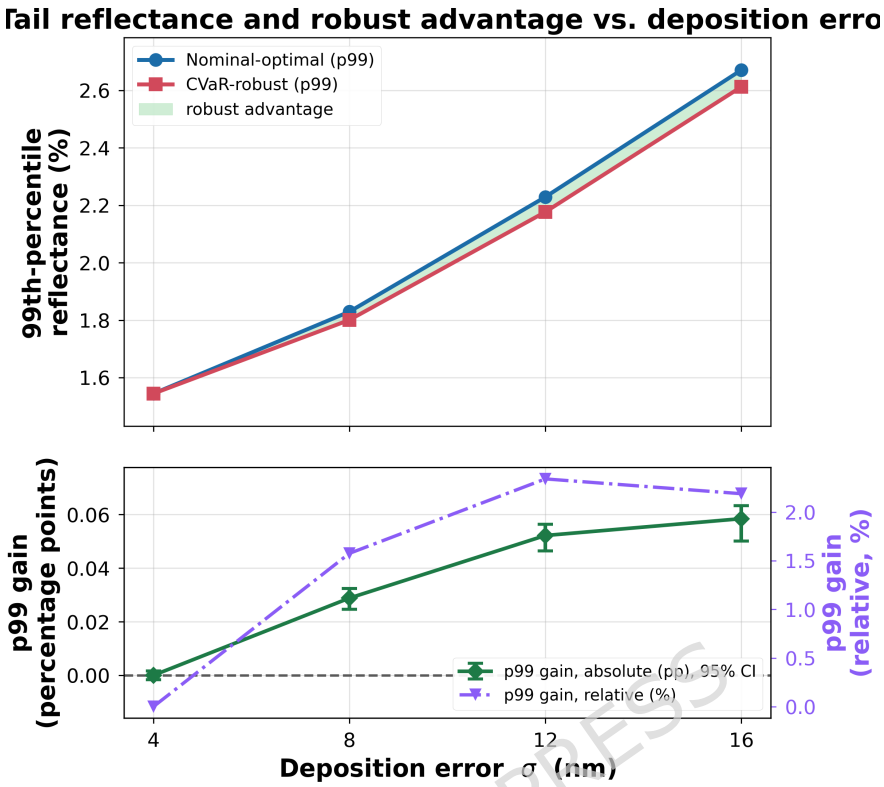


Figure 9: (a) 99th-percentile reflectance of both designs versus deposition error, with the shaded band marking the robust advantage. (b) The paired p99 reduction in percentage points with 95% bootstrap confidence intervals and an explicit zero line; positive values indicate that the robust design performs better. The advantage is absent at $\sigma_{\mathrm{dep}} = 4\,\mathrm{nm}$, opens above that break-even point, and then plateaus in relative terms.

3.9 Material-choice ablation

Three fixed six-layer recipes were optimized under the nominal objective on the same substrate and band: the working low-contrast MgF_2/SiO_2 stack, a variant with an MgF_2 top layer over alternating TiO_2/SiO_2, and a classical high-contrast TiO_2/SiO_2 stack. The best achievable band-mean reflectances are 1.439 %, 3.183 % and 12.774 % respectively (Fig. 10).

Within the design space tested – six layers, a fused-silica substrate, 430 nm to 780 nm, and these three fixed recipes – introducing the high-index material does not help; it raises the achievable reflectance by 1.74 pp in the mixed recipe and by an order of magnitude in the pure high-contrast stack. We state this deliberately narrowly. The result does *not* establish that high-index materials are unsuitable for anti-reflection coatings in general: high-index layers are standard in broadband and multi-band AR designs with more layers, different substrates, or optimized index profiles, and a fair comparison would re-optimize the layer count and ordering for each material system rather than holding them fixed. What the ablation does show is that in *this* low-contrast six-layer problem the optimizer derives no benefit from the high-index option. The mechanism is the familiar one: matching a substrate of index ≈ 1.46 to air favours a graded sequence of low indices, whereas a high-index layer

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introduces a large impedance discontinuity that the remaining layers must then compensate. We include the result because a negative outcome of this kind is a useful check that the pipeline is optimizing the intended objective rather than exploiting an artefact of it.

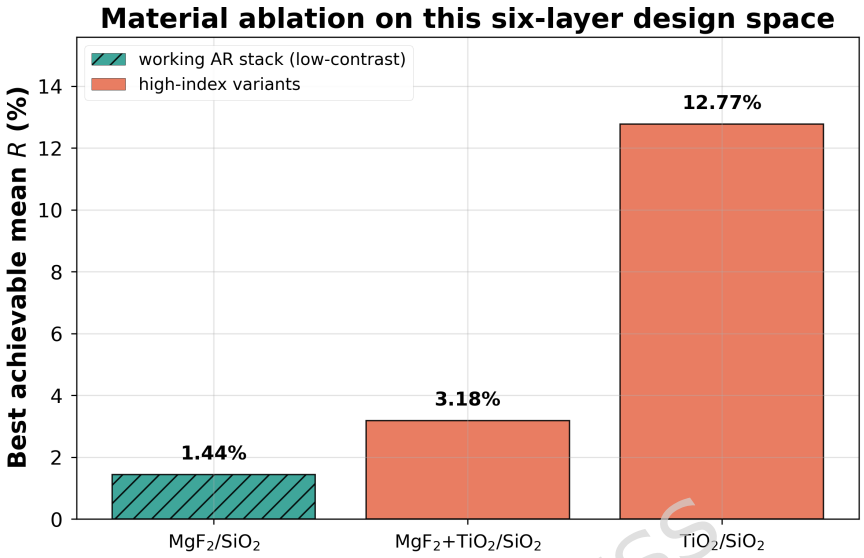


Figure 10: Best achievable band-mean reflectance for the three six-layer recipes tested here, on fused silica over 430 nm to 780 nm. The working MgF_2/SiO_2 stack is hatched. Within this design space, introducing TiO_2 degrades performance; this is a statement about the tested recipes, not about high-index materials in general.

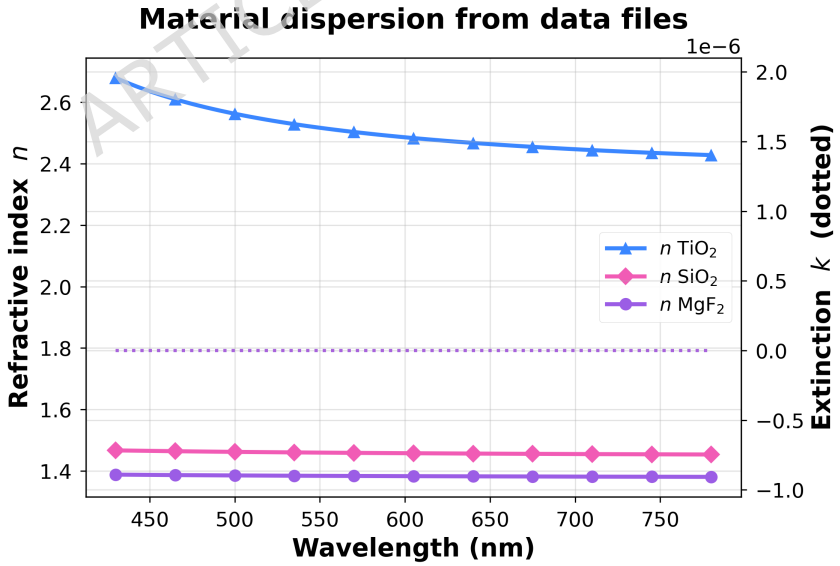


Figure 11: Tabulated refractive indices (solid, left axis) and extinction coefficients (dotted, right axis) of the three materials over the design band. All are transparent to the precision of the data.

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4 Discussion

4.1 What the method does and does not deliver

The central result is that a CVaR objective produces a design measurably more tolerant to deposition error than a nominally optimal one, at a small and quantified cost in ideal performance. The magnitude of that benefit is modest: 0.026 pp on the 99th percentile at $\sigma_{\mathrm{dep}} = 8\,\mathrm{nm}$, or 1.41 % in relative terms. We have taken care to establish this rather than assert it – through paired evaluation, bootstrap intervals and seed replication – because an improvement of this size is exactly the kind of claim that a single unreplicated workflow cannot support.

Whether such a gain is practically useful depends on the process. In a low-contrast MgF_2/SiO_2 system the reflectance landscape is shallow and there is little room between the nominal and the robust optimum. We would expect larger benefits in systems with sharper spectral features – narrowband filters, high-contrast stacks, or designs with many more layers – where the penalty for thickness error is correspondingly steeper. The break-even behaviour observed at $\sigma_{\mathrm{dep}} = 4\,\mathrm{nm}$ makes the operating condition explicit: robust design earns its cost only when the process error is large enough for the tail to matter.

4.2 Limitations

We list the limitations of this study explicitly.

Error model. The Gaussian, independent, uncorrelated error model of Eq. (4) is the study's single significant modelling assumption. Real deposition error is correlated between layers, may be systematically biased, and is generally non-Gaussian. A physically derived error distribution – from a growth model, or from measured process data – would make the CVaR objective correspondingly more meaningful, and could change the ranking of designs.

Magnitude of the effect. As discussed above, the improvements are small in absolute terms, and comparable in size to the seed-to-seed variation of the optimization itself. The consistency of the sign across seeds is what supports the conclusion, not the size of any single measurement.

Worst-case metric. The worst case is the maximum of a finite sample. It is reported for completeness but no conclusion rests on it, and it behaves irregularly across the sweep.

Scope of the ablation. The material ablation holds the layer count and ordering fixed. It is a statement about three specific recipes, not a general result about material selection.

Single material system. All results are for MgF_2/SiO_2 on fused silica over the visible band. Generalization to other systems is expected on physical grounds but is not demonstrated here.

Generator performance. The OT-CFM generator does not outperform direct CVaR optimization on the tail metric. Its contribution is design diversity, not a better optimum.

4.3 Outlook

The natural extensions are a physically derived deposition-error distribution; distributionally robust formulations that hedge against misspecification of that distribution itself [30, 31, 32];

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conditional generation, in which the target spectrum or the error magnitude becomes an input to the velocity field; and application to systems with sharper spectral features, where the tolerance penalty is steeper and the benefit of yield-aware design should be correspondingly larger.

5 Conclusion

We have moved fabrication tolerance from a post-hoc check into the design objective for multilayer anti-reflection coatings, minimizing the conditional value-at-risk of band-mean reflectance under deposition error through a differentiable transfer-matrix model, and coupling this objective to an optimal-transport flow-matching generator under a hard manufacturability constraint.

On a six-layer MgF_2/SiO_2 stack the robust design lowers the 99th-percentile reflectance by 0.0257 pp (95% CI $[+0.0230, +0.0274]$ pp; +1.41% relative) against a 0.011 pp nominal penalty, an improvement that reproduces across five independent optimization seeds. The advantage is absent at $\sigma_{\mathrm{dep}} = 4\,\mathrm{nm}$, opens above that break-even point, and then plateaus in relative terms while continuing to grow in absolute terms. A material-choice ablation documents that, within the six-layer low-contrast design space tested here, a high-index material degrades AR performance.

The gains are small, and we have reported them with the sampling uncertainty and the replication needed to distinguish them from noise. We regard the paired evaluation protocol as a contribution in its own right: robustness improvements in this domain are often of a magnitude that unpaired Monte-Carlo comparison cannot resolve, and reporting them without that machinery risks presenting sampling fluctuation as a design effect.

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Data availability

The code implementing the complete pipeline – the differentiable transfer-matrix solver, the CVaR objective, the OT-CFM generator, the paired evaluation and bootstrap machinery, and all figure-generation scripts – is publicly available at https://github.com/Erfan1379-hub/yield-aware-ar-coating. The optimized layer thicknesses of all designs reported here are given in Table 2 and are additionally released in machine-readable form, together with a complete hyperparameter record (Table 1) and a file containing every reported statistic with its confidence interval. The tabulated optical constants are from the cited public sources [39, 40]. All datasets generated during this study are contained in the repository.

Author contributions

E.G. is the sole author: he conceived the study, implemented the code, performed the simulations, analysed the results and wrote the manuscript.

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Competing interests

The author declares no competing interests.

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